

Applied Agentic AI for PMs/TPMs

Capstone Project

Mira: AI-Powered Project Intelligence Assistant

Company Profile

Nexora Pvt. Ltd. is a forward-thinking software development and IT services company headquartered in London, United Kingdom. Established in 2018, Nexora delivers cutting-edge digital solutions that help businesses across industries automate operations, improve decision-making, and scale efficiently. With a team of 120+ professionals, the company serves clients across Europe and Asia.

Nexora has recently won a contract to manage the AI Adoption Project for ABCDE Ltd., a mid-sized logistics and supply chain company. The project involves integrating AI-powered tools into core business functions including demand forecasting, route optimization, inventory management, and customer support over a 6-month timeline.

Your Role

You have been recently hired as an AI PM/TPM at Nexora Pvt. Ltd. Being a startup, almost everyone is overloaded with work and has to wear multiple hats. You work closely with data scientists, machine learning engineers, software developers, and product stakeholders to define project scopes, set milestones, and ensure timely delivery.

You have been onboarded to the ABCDE Ltd. AI Adoption Project and given access to the project description, timeline, risk assessment, and task board data.

Problem Statement

Nexora’s PMs/TPMs face significant operational challenges:

- Multiple projects running simultaneously, led by only a few team members
- Creating project plans from high-level descriptions takes 3-4 hours of manual work per project
- Risk assessments are done inconsistently — some projects have detailed risk matrices, others have none
- Weekly status reports take 1-2 hours to compile manually from task boards and meeting notes
- Missed milestones due to lack of proactive alerts — issues discovered only in retrospectives
- No standardized format for project plans, status reports, or risk documents across the team

The CTO has asked you to build Mira — an AI-powered project intelligence assistant that can automate project planning, risk assessment, and status reporting. Mira should take raw project data as input and generate structured, actionable outputs that PMs/TPMs can use immediately.

Curriculum Concepts You Will Apply

Concept	Mandatory / Extended	How It Applies in Mira
Multi-Agent System Design	Mandatory	Design agents for planning, risk assessment, status reporting, and milestone tracking

Orchestration Pattern	Mandatory	Select and document pattern for processing project data through specialized agents
Prompt Engineering	Mandatory	Write prompts that generate structured outputs grounded in actual project data, not generic templates
Data Grounding	Mandatory	Ground all outputs on the provided project files. Do not invent milestones, risks, or task statuses
Evaluation & Observability	Mandatory	Connect to a tracing tool, run the provided test dataset, document findings
Fine-Tuning	Extended (optional)	Train a model on corrected project plan or risk assessment examples
RAG / Knowledge Retrieval	Extended (optional)	Use project files as retrieval context for grounded generation
Human-in-the-Loop	Extended (optional)	PM/TPM reviews generated plan before sharing with stakeholders

Orchestration Pattern Requirements

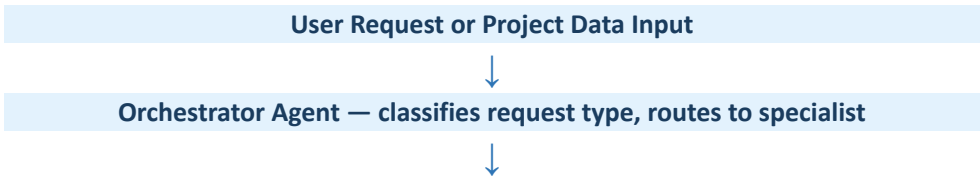
Your submission must document which orchestration pattern you selected and why.

Pattern	When to Use
Sequential Pipeline	Fixed-order: ingest project data → generate plan → assess risks → produce status report.
Router / Dispatcher	Route different request types (plan vs risk vs status) to specialist agents.
Hierarchical	Orchestrator breaks complex requests into subtasks for specialized agents.

Recommendation for Mira:
A Router pattern fits well because user requests vary: 'Generate a project plan' goes to the Planner Agent, 'What are the risks?' goes to the Risk Assessor, and 'Give me a status report for Sprint 3' goes to the Status Reporter. Alternatively, a Sequential Pipeline works if every input goes through all agents.

Suggested Architecture Overview

Reference architecture. Design your own and justify your decisions.



Planner Agent	Risk Assessor Agent	Status Reporter Agent
Ingests project description, generates plan with phases and milestones	Analyzes project data, generates categorized risk matrix	Reads task board, generates weekly status with blockers



Output: Structured Plan / Risk Matrix / Status Report

Possible Agents

Agent	Role
Orchestrator	Routes requests to the right specialist based on intent
Planner Agent	Ingests project description and timeline, generates structured project plan
Risk Assessor Agent	Analyzes project details, generates categorized risk matrix with mitigations
Status Reporter Agent	Reads task board data, generates formatted weekly status report
Milestone Tracker	Compares timeline vs current progress, flags at-risk milestones

You may combine, split, or rename agents.

What Mira Should Be Able to Do

Core Capabilities (Required)

Project Plan Generation: Ingest a high-level project description and generate a structured project plan with phases, key activities, milestones, deliverables, and timeline estimates. Must be grounded in the actual project description.

Risk Assessment: Analyze project details and generate a categorized risk matrix with risk description, potential impact, and mitigation strategies. Risks must be relevant to the specific project.

Weekly Status Report: Ingest task board data (CSV or API) and generate a formatted weekly status report showing: tasks completed, in progress, blocked items, upcoming milestones, and overall health.

Extended Capabilities (Choose at least one)

Milestone Alert System: Compare timeline against current progress and flag at-risk milestones with reasons and suggested actions.

Stakeholder Update Generator: Generate executive-friendly updates tailored for different audiences (leadership vs engineering vs client).

Resource Allocation Advisor: Based on task assignments, suggest whether the team is over/under-allocated.

Competitive/Market Analysis: Given a project domain, research and generate a competitive landscape analysis.

Evaluation Requirement

Connect your Mira workflow(s) to any evaluation/observability tool. Run the baseline test dataset below and document results.

Baseline Test Dataset for Core Capabilities

Run each input. Check outputs against the expected criteria.

ID	Test Input	Type	Expected Output Must Contain
T1	Generate a project plan for the AI Adoption Project at ABCDE Ltd. Use the provided project description.	Plan (detailed)	Must include phases, milestones, timeline. Must be grounded in the actual project description. Must NOT invent goals not in the description.
T2	Generate a project plan for: 'We want to build a chatbot.'	Plan (vague)	Must flag as insufficient detail. Must ask for scope, timeline, team size. Must NOT generate a full plan from one sentence.
T3	Generate a risk assessment for the AI Adoption Project at ABCDE Ltd.	Risk (detailed)	Must produce categorized risks relevant to logistics/AI. Must NOT include generic risks unrelated to project.
T4	Generate a risk assessment for: 'New project starting soon.'	Risk (vague)	Must flag as insufficient. Must ask for project details. Must NOT invent project-specific risks from nothing.
T5	Generate a weekly status report using the sample task board for Sprint 3.	Status (data)	Must show tasks by status. Must reference actual task names from CSV. Must NOT invent tasks.
T6	Generate a weekly status report for: 'Things are going fine.'	Status (no data)	Must state no task data available. Must NOT fabricate task statuses or progress percentages.
T7	What are the top 3 risks for the ABCDE Ltd project?	Risk analysis	Must reference actual risks from risk CSV. Must NOT invent risks not in the data.
T8	Which tasks are blocked or at risk of missing their deadline?	Tracking	Must identify T024 (Security review - BLOCKED). Must check due dates against current progress.
T9	Generate a project plan for a 2-week project with no other details.	Plan (edge)	Must ask for scope, goals, deliverables. Must NOT generate detailed plan from timeline alone.
T10	Summarize current project status: tasks done, in progress, to do, blocked?	Status summary	Must count correctly from task board CSV. Counts must match actual data.

T11	What milestones are coming up in the next 2 weeks?	Timeline	Must reference actual milestones from project_timeline.csv. Must NOT invent milestones.
T12	Generate a stakeholder update email for Sprint 2 progress.	Comms	Must reference Sprint 2 tasks. Must be professional. Must NOT include Sprint 3+ tasks.

Fine-Tuning (Optional Enhancement):
Fine-tune the Planner or Risk Assessor using corrected examples. Not required but recognized in scoring.

Inputs Provided

Input File	Description
project_description.txt	High-level description of the ABCDE Ltd. AI Adoption Project (scope, goals, approach)
project_timeline.csv	6-month project timeline with 8 phases, activities, and milestones
project_risks.csv	10 categorized risks with impact descriptions and mitigation strategies
sample_task_board.csv	25-task Kanban board export with statuses, assignees, sprints, and due dates
eval_mira_inputs.txt	Baseline test inputs (the table above in text format)

About the sample data:
The task board has a mix: 3 Done, 3 In Progress, 1 Blocked (T024 - Security review), and the rest To Do. The Blocked task is intentional — test whether Mira correctly identifies it.

Tools and Technologies

Category	What You Need	Examples (not exhaustive)
Workflow/Agent Platform	Low-code or no-code platform	n8n, LangFlow, or any platform
LLM	Reasoning, planning, generation	OpenAI (gpt-4o-mini), Claude, Gemini
Data Ingestion	Read/parse CSVs, text files, PDFs	Built-in file nodes, HTTP Request, code
Observability	Monitor traces, track quality	LangWatch, Langfuse, or any tool
Output/Export	Structure outputs for team use	Markdown, Google Docs, email, Slack
Task Board API (optional)	Read task data programmatically	Trello API, Jira API, or CSV upload
Authentication	Secure access	OAuth2, API keys

Assignments

Q1: Ideation (15 points)

Identify 2-3 use cases where AI agents can increase PM/TPM productivity at Nexora. For each: pain point, agentic solution, inputs/outputs, 3 success metrics, and knowledge base needed.

Q2: Build Mira (45 points)

Build Mira with Design/Architecture (diagram, orchestration, tools, agents), Implementation (core + extended, observability, baseline testing), Cost/Evaluation (cost per plan, 3+ metrics, observations). 1-2 page writeup.

Sub-criteria	Points
Design: architecture + orchestration + tool rationale	10
Core capabilities end-to-end	12
Extended capability	8
Observability connected	5
Baseline dataset tested with results	5
Cost analysis and evaluation strategy	5

Q3: Program Charter (25 points)

Program charter for Mira: vision, scope, success criteria, timeline, risks, stakeholders, decision-making, rollout plan.

Q4: Reflection and Next Steps (15 points)

1-page reflection: trace findings, improvement plan, evaluation/fine-tuning connection, privacy considerations, rollout for 10 PMs/TPMs.

Submission Guidelines

- Export workflow(s) as JSON, upload to GitHub with README
- Screenshots: workflow canvas, Mira in action, observability traces
- Architecture diagram, baseline test results documented (all 12 inputs)
- Submit GitHub repository link along with documents